

A Review: Contrast-Limited Adaptive Histogram Equalization (CLAHE) methods to help the application of face recognition

Purnawarman Musa
Faculty of Computer Science and
Information Technology
Universitas Gunadarma, Indonesia
Email: p_musa@staff.gunadarma.ac.id

Farid Al Rafi
Faculty of Computer Science and
Information Technology
Universitas Gunadarma, Indonesia
Email: faridalrafi@gmail.com

Missa Lamsani
Faculty of Computer Science and
Information Technology
Universitas Gunadarma, Indonesia
Email: missa@staff.gunadarma.ac.id

Abstract—Image processing technology in hardware and software has a significant development in all aspects of human civilization. Image processing is growing rapidly in various fields such as medical image, Biometrics, Object identification, Object character recognition (OCR), and many more. One biometric technique in an application detects and recognizes faces. Objectives of face recognition include recognizing emotions, identifying offenders, applying to security systems, image processing to access a door as an application of computer and human interaction. The facial recognition system in this research was conducted live streaming using Local Binary Pattern Histogram (LBPH) method. There is a constraint to a light intake received by the camera through a sensor that captures the image.

Based on the results of our research to make better facial recognition, we propose using adaptation methods on the intensity of light that is around the image capture object. The method of adaptation in this research is the Contrast Adaptive Limited Histogram (CLAHE).

Index Terms—Image Processing, Biometric, Contrast-Limited Adaptive Histogram (CLAHE), Local Binary Pattern Histogram (LBPH), Face Recognition

I. INTRODUCTION

Now, in the world of information technology, biometrics are relevant to the technology used for physical analysis and human behavior for authentication. such as the introduction of the physical human with the introduction of fingerprints, retina, iris, facial patterns, and signatures.

Face recognition support to human beings perform automatically any times, every day and practice with like easy task, but the process in the computer it has implemented to be a consisting of many different and connected parts of the task. The computer needs dataset as variables that can the quality of the methods, for example, version of illumination, low level of the resolution, occlusion, etc.

The use of information technology in biometric tagging can be used in the identification model, where the biometric system identifies each person and searches through the same code in the database that has been compiled. The use of biometric is considered to have a higher level of security than traditional systems that use passwords or ID Card for authentication on accessing a system [1].

Usually, the task of recognizing in computer science for face recognition a someone based on identically from facial image. In the last decades, two method face recognition become very popular, the first is methods developed and the second are the high quality of the current videos/cameras.

In the computer vision, the research of prototypes integrates action with the result the retina analog functions (Smart Image Sensor using analog Hmax) capable of realized autonomous vision systems accompanied by image treatments derived from using $0.35\mu\text{m}$ CMOS technology as measurement and tools an image sensors. In the computer as autonomous vision systems, the action of mathematical calculation doesnt need a high precision and an analog filtering approach. Their project proposal of prototypes integrates to embed a system on the retina analog functions into the image sensors. The superior position is projected in the instance of comparing with series of the digits 0 and 1 plan into effect is to specified condition an approved of speed determination and power consumption [2].

Adaptive Histogram Equalization (AHE) tends to over amplify noise in the relative region of an image. A variant of Adaptive Histogram also Contrast-Limited Adaptive Histogram Equalization (CLAHE) aims to prevent over amplification by limiting contrast amplification. The amplification contrast around the given pixel value is given by the slope function transformation. CLAHE limits present the amplification by clipping based a predefined value in the histogram before CDF computing (cumulative distribution function) [3].

II. RELATED WORK

A. Image Processing

The introduction of the future using digital image processing as a basic concept and implementation for digital images is the development phase of the field of study and the discipline of image processing, e.g: basic of images, compilation of images in relating to spatial and frequency an area of image, restoration, color unbalance, wavelets, image compression, morphology operation, segmentation image, motion and detection objects, description of images, and the basics of objects

and patterns recognition [4].

Expanded to development an image processing or novelty in area image processing has coverage about deep learning and deep neural networks, convolutional method, Neural Networks, the Scale-Invariant Feature Transform (SIFT), Maximally-Stable Extremal Regions (MSERs), k-means clustering and etc [5].

However, the image processing to an execution of these rules to be followed in calculations is usually carried out in digital form (using the computer) and need for a particular purpose intensive calculations not really compatible with the restriction of embedded systems.

The embedded systems of camera sensors have the constraints of the quality of the images, a speed of treatment, the performance system of embedded processing, and low power consumption [6].

B. Biometric

Before applied to computer technology, access systems on a device still use special objects such as access cards or use memory power and the blocks are core low-level building a multi-biometric system is implemented [7]. There are two reasons for using Biometric data today:

- 1) As a verification tools, where the required data is on an object (such as a document or credit card). But they can be duplicated so that it can be misused by someone else.
- 2) As a verification codes, usually using a password. Sometimes we forget the codes. Even if we use the best encryption algorithm, there are still algorithms that can do the decryption

The use of biometrics has several advantages, as follows:

- Can not be lost or forgotten
- Difficult to duplicate, share or change hands
- Authenticity is guaranteed

C. Contrast-Limited Adaptive Histogram Equalization (CLAHE)

Contrast-Limited Adaptive Histogram Equalization is an image processing technique been used to make greater in the contrast of an image. CLAHE operates in a small area in an image, known as a tile call, rather than an entire image. The application of the histogram is separate from the histogram in general, where the adaptive method is to calculate the function of the state of being contrast color to change every tile. Each tile contrast increases the histogram of the result on region approximation to it by the distributed value. The neighboring tiles then merged to avoid induced boundaries. The contrast, mainly in homogeneous areas, will be limited to avoid amplifying a noise that could be present in the image.

In figure 1, the Histogram Equalization is gray on the image x and the value i is a gray level amount to a maximum value of 256. Whereas to look for Probabilities at pixel image from level i in an image to calculated as follows;

$$P_x(i) = P_{(x=i)} = \frac{n_i}{n}, \Rightarrow 0 \leq i < L \quad (1)$$

which,

- The variable L is the whole a number of gray levels present at the image (usually 256),
- The variable n is the number of pixels from dataset the image, and
- The variable $P_x(i)$ a representation of the image histogram for the pixel into an index to i

Algorithm 1 Histograms Equalization Function

Require: Image_{Input} is Image_{ORIGINAL}, which Image_{Input} is convert to grayscale image

Ensure: Image_{Output} is Image Histogram Equalized from Image_{Input}

- 1: Using the function to read the image
 - 2: Find the result into variable equ with convert image to histogram equalize
 - 3: Stacking from the images side-by-side
 - 4: Show the resulting image
-

When bringing with one quality image with distinct sources of light dynamics conditions, the make the same in quantity and find to check the results.

Histogram equalization function has an approved of the result during the histogram of the image outcome is confined to an intense region. It inoperable places where the relatively great size intensity difference and histogram covering an area, which used to refer to two reflecting a lot of light and no light as representation gray level pixels are present in figure 1.



(a) Original Image without AHE

(b) Result Image with AHE

Fig. 1. Result image use AHE method

The histogram equalization calculates the global contrast from the image. For a few cases, the image show unclearness result. For example, this images shows an input image (in figure 2(a)) and the figure 2(b) is result after global histogram equalization.

The scenery behind the main object of contrast has become better after the result of the image for histogram equalization. The form the comparative and superlative of the face of the statue in both images (between in fig. 2(a) and fig. 2(b)). Raw data for image is lost of the information as an image, if due to over-brightness.

From this problem, adaptive histogram equalization is used. In this section, the image is divided into small blocks called tiles (tile size 8×8). This tiles used to refer to every one of

Algorithm 2 Contrast-Limited Adaptive Histogram Equalization (CLAHE) Function

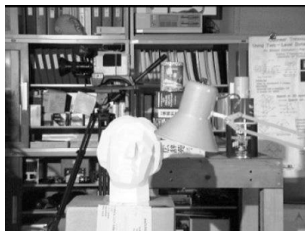
Require: Image_{Input} is Image_{ORIGINAL}

Ensure: Image_{Output} is Image Contrast-Limited Adaptive Histogram Equalization (CLAHE)

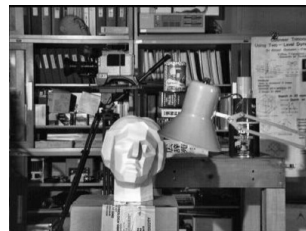
- 1: Find and read the image to put into variable img
 - 2: Call function of CLAHE with clipLimit=2.0 and matric = 8×8
 - 3: The result CLAHE into variable cll
 - 4: Show the result CLAHE image
-



(a) Original Image as image test



(b) Result Image using AHE



(c) Result Image using CLAHE

Fig. 2. CLAHE method testing

two or more of these blocks is histogram equalized as usual. The results that in a small area becomes the histogram would be limited to a small region. If the noise is there, it will be amplified. To render avoid the noise, contrast limiting is relevant. If any histogram bins are above the specified contrast limit, those pixels are cut (clipped) and distributed uniformly to other bins before applying histogram equalization. After equalization, to remove artifacts in tile borders, bilinear interpolation is applied. See the result of the image (in figure 2) and compare it with results of image (in figure 2(c)), especially the statue region.

D. Face Recognition Algorithm

The face recognition method is used to get geometric features of a face to recognize a human face. The most famous algorithm is 3 (three) Face recognition algorithms that can be used include:

a) *Eigenfaces*: is one model of face recognition algorithm that uses eigenvector. The inventor of the eigenface

algorithm in facial recognition was developed by Sirovich and Kirby (1987) and published by Matthew Turk and Alex Pentland in face classification (Turk, Pentland 2002). Classification is obtained by comparing face data from existing face database [8].

b) *Fisherfaces*: is a facial recognition algorithm developed using Principal Component Analysis (PCA) which is also the core of the Eigenface method, and is found by statistician Sir R. A. Fisher (Fisher, 1936). The Fisherfaces method uses the class-specific transformation matrix, so they do not capture illumination as obviously as the Eigenfaces method. It's important to mention, that the performance of the Fisherfaces heavily depends on the input data as well [9].

c) *Local Binary Pattern Histogram (LBPH)*: is a facial recognition technique with a method of extracting local features from images [10]. LBPH was first introduced in 1994. Since it was discovered, it has become a good technique in terms of texture classification.

The spatially enhanced Local Binary Pattern (LBP) and Local Binary Pattern Histogram (LBPH) a system of methods used in a particular area has succeed in achieving an established position in the field of Face Recognition (FR) and obtain something from many face analysis approaches [11].

E. LBPH in face recognition

Local Binary Pattern Histogram (LBPH) is one of the applied face recognition using binary pattern. This algorithm looks like a very simple but very effective and efficient way to describe the texture and shape of a digital image. LBPH has simpler computing, making it a popular approach in various applications [12].

The LBPH method is operated by dividing an image into a small regional. The introduction of the operator's texture to the pixel label of a image with the environment of each pixel and represents the result on a number binary.

The concept of using LBP is simply described as follows:

- 1) Divide the image to be used into multiple matrix (example divided by 16 matrices)
- 2) Each pixel present in a matrix will be compared with pixels from the 8 neighbors (top left, middle left, lower left, right top, and so on).
- 3) If the middle value is greater than the pixel value then give the value "0" and if the middle value is smaller than the pixel value it will give value "1". Then we will get an 8 digit value of binary value.
- 4) Calculate the value of histogram, matrix, frequency of each number obtained (ie, any combination of which is smaller and larger than the center. The value of this histogram is obtained in 256 vector dimensions.
- 5) Using Normalization function at histogram value.
- 6) Sync the histogram value of each matrix, so it will give the vector feature value of the whole image.
- 7) In face recognition, the captured value will be compared to the histogram value, if close to the same ideal value, then the face will be recognized as the user.

III. METHODS AND IMPLEMENTATION

In the systematic investigation into and study our project, we discuss an image contrast enhancement algorithm specifying the image overcomes limitations in standard Histogram Equalization (HE). In preference to implementing our research for contrast enhancement, there are two primary features. The first feature is called Adaptive Histogram Equalization (AHE), which a wide divergence between two groups the images into regions and performs local HE, and the second feature is a specified name Contrast-Limited Histogram Equalization (CLAHE), which reduces noise by partially reducing the local Histogram Equalization. Bilinear interpolation is used to avoid visibility of region boundaries.

A. Experiment of Contrast enhancement limit

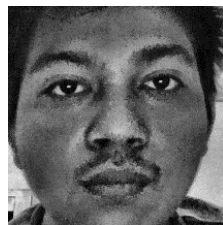
Introducing the Contrast enhancement limit, an activity identifies clearly and definitely like actually existing a scalar quantity in the range $[0, 1]$. The resulting values are higher than the limit, then we need to add more contrast. The function Contrast Limit features a contrast that contributes to a result that keeps oversaturation of the image. These areas are typical by a high peak in the histogram of the individual item image tile due to many pixels grading in the same gray level (see fig. 3).



(a) ClipLimit=0.5



(b) ClipLimit=1.0



(c) ClipLimit=2.0

Fig. 3. Compare of contrast enhancement limit

Without the clipped limit being the case that, it could things that have been the result in some cases it less good the original image.

B. Experiment of CLAHE method with illumination condition

Based on the CLAHE algorithm shown in alg. 3, then the first CLAHE function Read the indexed color image as Original Image. After the CLAHE function is executed, it will get as shown in figure 4.

Algorithm 3 Contrast-Limited Adaptive Histogram Equalization (CLAHE)

Require: Read the indexed color image as Original Image

Ensure: Showing the image result into the RGB color

- 1: Cause to change in form the listing image in an index (each pixel) expressing an act a true-color (RGB) image
- 2: Become different the RGB image using function change into the $L^*a^*b^*$ color space
- 3: Specified scales from values to the area of variation between upper and lower limits on expected by the Adaptive Histogram Equalization (AHE) function
- 4: Present CLAHE method on the L a part of channel using **Number of tiles** and **Contrast enhancement limit**
- 5: Plotting the scale the result to return on the range the state of being used for purpose $L^*a^*b^*$ color space
- 6: To change the following as the consequence image back into the RGB color space

In this research, we conducted experiments in same places, but different illumination to prove the application of CLAHE obtained clear image results in dark conditions or bright conditions, as shown in Figure 4.



(a) Dark conditions



(b) Bright conditions

Fig. 4. The result of Image using CLAHE method with different illumination

C. Experiment of Local Binary Patterns Histogram approach for Face Recognition

The implementation of LBPH by every one of an image must be analyzed to doing independently. The LBPH method is a moderate extent simpler to analyze, in the perceived by a sense that characterizes each image in the dataset locally; and on which a newly cannot describe and provide as an image, the LBPH is better method analysis on face recognize and present the different from the result to each of the datasets images. This method has provided analyze to the images the local patterns in each position in the image.

For detecting a face from an image, the method will return the x and y coordinates of used to refer to the whole a faces in a dataset. The process of thinking, that application can detect only facial recognize. A human being regarded as the tester a recognizer has the definite about face recognition, using the

LBP method to recognize a specific person will be given a vector of faces belong to the person.

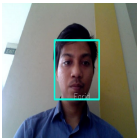
For this, We propose the following algorithm 4:

Algorithm 4 LPBH for Face Recognition

- 1: Read the input video for the next frame
- 2: The classification the faces from detect all in frame
- 3: Recognize each faces for the detected faces
- 4: The actions are successful, produce a green rectangle
- 5: Write the accuracy with the level of each face in the frame to a window
- 6: Repeat steps 1-5, until no more frames in the input video

In instruction to test the application using the training set from dataset images. For example, looking to analysis, and that the results have discerned visually illustrating a general of detection process and process of recognition of faces. The theory the back side is quite interesting (see table I).

TABLE I
IMPLEMENTATION OF CLAHE FOR FACE RECOGNITION

Case	Face Recognition	Description of the Image
1		Capture Image: 1. Exterior Room 2. Lumen = 240 lx 3. Accuration Recognition = 75% 4. Face Detect founded 5. Face Recognition founded
2		Capture Image: 1. Exterior Room 2. Lumen = 198 lx 3. Accuration Recognition = 85% 4. Face Detect founded 5. Face Recognition founded
3		Capture Image: 1. Interior Room 2. Lumen = 98 lx 3. Accuration Recognition = 95% 4. Face Detect founded 5. Face Recognition founded
4		Capture Image: 1. Interior Room 2. Lumen = 65 lx 3. Accuration Recognition = 50% 4. Face Detect founded 5. Face Recognition founded
5		Capture Image: 1. Interior Room 2. Lumen = 65 lx 3. Accuration Recognition = 95% 4. Face Detect founded 5. Face Recognition founded

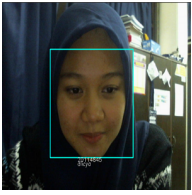
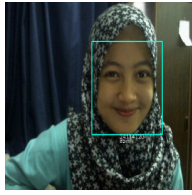
Experiments in the image results of the CLAHE test above in table I, there is outdoor testing and indoors that result in different lumens. Each the result can recognize the face of the user tester, where every previous face has been stored in a database. There are also tests on twins that have been tested in cases 4 and 5. Although the success to detect and recognize the face of one of the twins only succeeded 50%, the results

in twins can detect and recognize their faces. The author has obtained permission from the people in the picture.

D. Experiment of Detection and Recognition with Two Face per Frame

From the test result in table II, this experiment test intend to detect and recognize every face in one frame. The first user was tested whether it can detect and recognize the face of the user. Then, testing on the second user was tested to detect and recognize the face of a second user. The final test of this experiment is testing by capturing images that contain more than one face in the image frame. From the results in the 3rd column in table II, the face detection and facial recognition experiment for one frame is more than one face successfully recognized and applied to our research.

TABLE II
DETECTION AND RECOGNITION TWO FACE FOR ONE FRAME

Testing User 1	Testing User 2	Testing Both Users
		
		

IV. CONCLUSION

From the improved image results based on the contrast boundary that has been adapted HE, it can improve the image obtained. If the contrast that captured by its image is very dark, the contrast value will be brightening on the image result. Conversely, if the image captured is very bright, then the contrast value will be reducing on the image result and become darker.

The application of LBP method for face recognition on this test results have been done, where this research can show detection and face recognition successfully recognize the face of every user who is outside and indoor received through the camera. But when testing twins, sometimes it resulting error when doing face detection and recognition incorrect face matching.

The continuation of this research for appropriate future technologies, as biometric data for the authentication of a system. Examples of applications that can be done are the attendance system and automatic door access by using a recognizable user face.

V. ACKNOWLEDGEMENT

This research was fully given assistance to realize this project by the Department of Computer Science. The authors thankfully acknowledge Universitas Gunadarma for especially financially make available for use research funding and authorization to access laboratory in using the research facilities.

REFERENCES

- [1] P. Musa, N. Yuliani, and M. Lamsani, "Rancang bangun pengendali pintu otomatis berdasarkan ciri wajah menggunakan metode euclidean distance dan fuzzy c-mean," 2008.
- [2] M. Paindavoine, J. Dubois, and P. Musa, "Neuro-inspired smart image sensor: analog hmax implementation," vol. 9403, 2015, pp. 9403 – 9403 – 8. [Online]. Available: <https://doi.org/10.1117/12.2079093>
- [3] S. M. Pizer, E. P. Amburn, J. D. Austin, R. Cromartie, A. Geselowitz, T. Greer, B. ter Haar Romeny, J. B. Zimmerman, and K. Zuiderveld, "Adaptive histogram equalization and its variations," *Computer Vision, Graphics, and Image Processing*, vol. 39, no. 3, pp. 355 – 368, 1987.
- [4] A. Rosebrock, *Practical Python and OpenCV: An Introductory, Example Driven Guide to Image Processing and Computer Vision*. PyImageSearch, 2014.
- [5] R. C. Gonzalez and R. E. Woods, *Digital image processing*, 4th ed. Upper Saddle River, N.J.: Prentice Hall.
- [6] P. Musa, S. A. Sudiro, E. P. Wibowo, S. Harmanto, and M. Paindavoine, "Design and implementation of non-linear image processing functions for CMOS image sensor," in *Optoelectronic Imaging and Multimedia Technology II*, vol. 8558, 2012, p. 85580O.
- [7] R. I. Hammoud, B. R. Abidi, and M. A. Abidi, *Face Biometrics for Personal Identification: Multi-Sensory Multi-Modal Systems*. Springer, 2007, p. 275.
- [8] A. L. Ramadhani, P. Musa, and E. P. Wibowo, "Human face recognition application using pca and eigenface approach," in *2017 Second International Conference on Informatics and Computing (ICIC)*, 1-3, pp. 1–5.
- [9] P. N. Belhumeur, J. P. Hespanha, and D. J. Kriegman, "Eigenfaces vs. fisherfaces: Recognition using class specific linear projection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 19, no. 7, pp. 711–720, Jul. 1997.
- [10] H. Dienillah, "Aplikasi pengenalan wajah menggunakan algoritma lbph berbasis opencv," 2014.
- [11] B. Yang and S. Chen, "A comparative study on local binary pattern (lbp) based face recognition: Lbp histogram versus lbp image," *Neuro-computing*, vol. 120, pp. 365 – 379, 2013, image Feature Detection and Description.
- [12] K. Salton, "Face recognition: Understanding lbph algorithm," Towards Data Science, 2017.